Why GitHub is the backbone of blockchain development

There's a significant level of transparency and co-operation required to be built on top of a blockchain compared to traditional software systems; however, once deployed on a blockchain, the code is immutable (which means that if there's an error in it or a vulnerability, the consequences cannot be corrected). Therefore, the stakes are high and it's critical to use a development platform that allows for structured workflows, ongoing validation of the code and global peer oversight. GitHub provides the tools necessary to meet these requirements by being an effective version control system, enabling developers to view and track every change made to the codebase. In addition, creating branches in which to explore or develop new functionality, reviewing all the differences between versions of the code and reverting to the previous version when needed, helps ensure that the final product is stable and ready for deployment.

A key aspect of GitHub's alignment with the fundamental values of blockchain is transparency. Everyone can see all the public repositories. Therefore, users (and any other stakeholders) can independently verify how each application operates (rather than relying on an opaque implementation), suggest modifications, and identify any vulnerabilities (e.g., bugs) that need to be remedied. This creates trust between users and within the wider community.

Another component of GitHub's contribution to blockchain is the enabling of global collaboration. Developers are (asynchronously) contributing to projects from all around the world, and they have continuous access to these projects regardless of what time zone they are in. This is similar in principle to the decentralised governance model of DAOs (decentralised autonomous organisations) wherein both the decision-making and the developmental effort are shared/distributed among all network participants rather than centralised in one entity.

GitHub provides many development tools. One of these is GitHub Actions, which can create automated testing and deployment pipelines for developers. These tools can be extremely beneficial for developers to validate their smart contracts during the testing phase (prior to real world deployment), significantly reducing the risk of making costly mistakes. GitHub supports the development of products and acts as a pre-validation layer where developers can confidently test the logic of their smart contracts before these become part of the immutable blockchain.

GitHub workflow in blockchain projects

The immutable code deployed on a blockchain makes it imperative that blockchain projects follow a disciplined development process. Once a smart contract is deployed on the blockchain network, it cannot be changed easily; therefore, projects must perform thorough validations and collaborate with each other prior to deployment.

GitHub provides a structure that supports the development, review and deployment of blockchain projects. A typical blockchain project hosted on GitHub adheres to a defined repository structure that separates the components of the overall project. For example, smart contracts are maintained in their own directory, while front-end user interfaces, configuration scripts and documentation are organised into separate modules. This leads to clarity of project components, ease of maintainability, and efficiency of collaboration among distributed teams when developing decentralised applications.

Version control system (VCS) branching is a vital mechanism for codebases in the blockchain. Branches are created for new functionality or to address error/malfunction issues without altering the current coding structure. Branching allows developers to create new functions and correct errors away from the live smart contract, so that there is an isolated area for functionality development as well as experimental results until development is complete. Upon completing development, developers submit their change for approval through a process called a pull request. This allows for collaborative validation of the proposed changes before they are merged into the main codebase.

Because of the collaborative validation of pull requests, contributors can find and fix any logical errors within their code before it is deployed. Due to the significant financial and security implications of smart contracts, developers must conduct more thorough reviews than are needed for traditional software applications. GitHub provides several tools to assist with reviewing code, including in-line comments, threaded discussions, and approval processes. These help ensure that only the highest quality codes are accepted into the main codebase.

Integrating CI/CD into the blockchain development process allows developers to use automated systems to test and validate their code changes to instantiate and execute smart contracts. GitHub Actions provides a way to automate the execution of smart contract testing using frameworks so that each code change can be checked against pre-established conditions. Automating testing through CI/CD pipelines mitigates human error and helps ensure consistent results before deploying to the underlying blockchain networks.

The GitHub workflow creates a controlled environment for blockchain logic, providing a framework to test, refine and validate the code before it is deployed to production.